COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	**CPE212 / Automating Server Management**	**SECTION:**	**CPE31S2**
DATE PERFORMED:	**Sept 17 2026**	**DATE SUBMITTED:**	**Sept 19 2026**
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ACTIVITY NO. 8.1 **Install, Configure, and Manage Availability Monitoring tools**

1. PROCEDURE OUTPUT

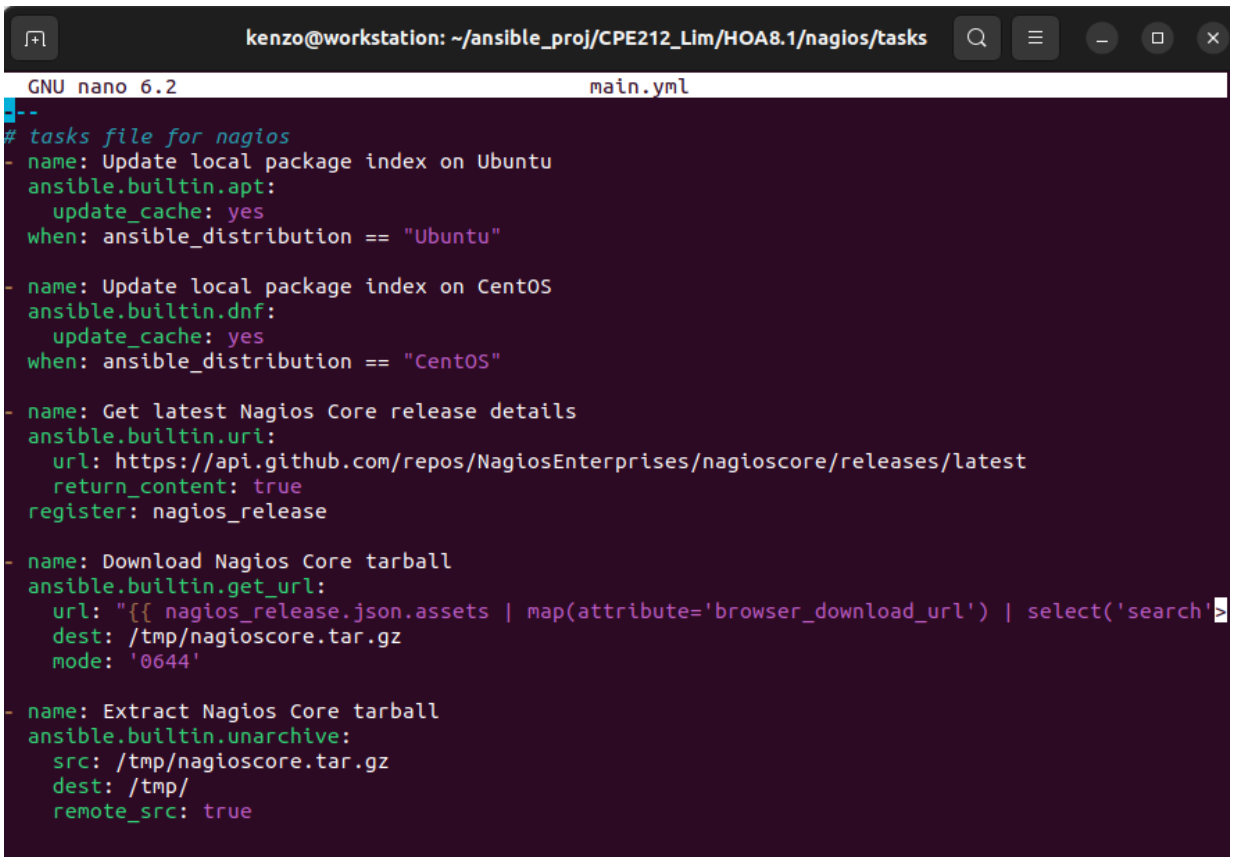
1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity.

Step 1: Created Nagios roles and configured tasks main.yml

```
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA8.1$ ansible-galaxy init nagios
- Role nagios was created successfully
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA8.1$ ls
install_nagios.yaml  inventory.yaml  nagios
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA8.1$ cd nagios
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA8.1/nagios$ ls
defaults  files  handlers  meta  README.md  tasks  templates  tests  vars
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA8.1/nagios$ cd tasks
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA8.1/nagios/tasks$ ls
main.yml
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA8.1/nagios/tasks$ nano main.yml
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA8.1/nagios/tasks$
```

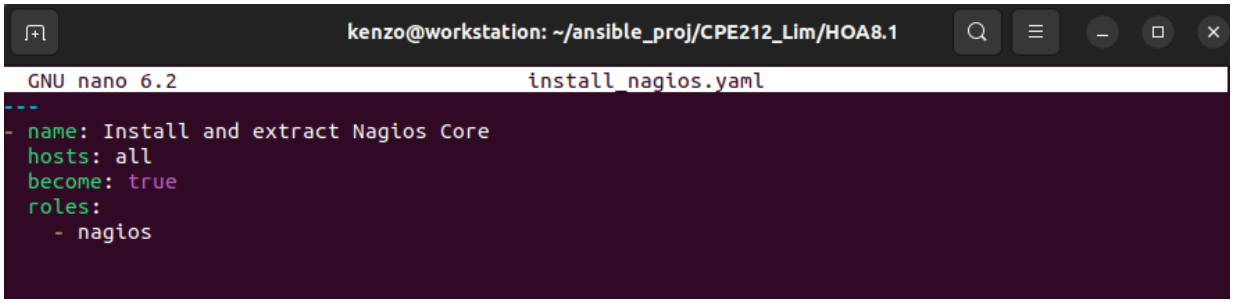
I ran ansible-galaxy init nagios to automatically generate the standard role folder structure, including the tasks, handlers, defaults, and vars directories. The terminal confirmed with "Role nagios was created successfully," and I then navigated into nagios/tasks to begin configuring.

Step 2: Configuring main.yml of nagios role



Using nano main.yml, I placed all five tasks inside roles/nagios/tasks/main.yml: updating the package index on Ubuntu and CentOS (each guarded by a when condition), fetching the latest Nagios Core release info from GitHub, downloading the tarball, and extracting it. The when conditionals ensure only the matching OS task runs, which is what makes the same role work on both distributions.

Step 3: Creating a playbook



I created install_nagios.yaml in the project root, which simply calls the nagios role under the roles: section with become: true.

Step 4: Review of the task logic (Get / Download / Extract)

```
- name: Get latest Nagios Core release details
  ansible.builtin.uri:
    url: https://api.github.com/repos/NagiosEnterprises/nagioscore/releases/latest
    return_content: true
    register: nagios_release

- name: Download Nagios Core tarball
  ansible.builtin.get_url:
    url: "{{ nagios_release.json.assets | map(attribute='browser_download_url') | select('search',>
    dest: /tmp/nagioscore.tar.gz
    mode: '0644'

- name: Extract Nagios Core tarball
  ansible.builtin.unarchive:
    src: /tmp/nagioscore.tar.gz
    dest: /tmp/
    remote_src: true
```

The Get task queries GitHub's API and stores the release info as nagios_release, the Download task filters that data for the .tar.gz link and saves it to /tmp/nagioscore.tar.gz, and the Extract task unpacks it into /tmp/ using remote_src: true since the file already exists on the target host..

Step 5: Executing the playbook

(without roles)

```
PLAY [Install nagios on Ubuntu and CentOS] *************************************************

TASK [Gathering Facts] *********************************************************************
[WARNING]: Platform linux on host server1 is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [Update local package index on Ubuntu] ************************************************
changed: [server1]

TASK [Update local package index on CentOS] ************************************************
skipping: [server1]

TASK [Get latest Nagios Core release details] **********************************************
ok: [server1]

TASK [Download Nagios Core tarball] ********************************************************
changed: [server1]

TASK [Extract Nagios Core tarball] *********************************************************
changed: [server1]

PLAY RECAP *********************************************************************************
server1                    : ok=5    changed=3    unreachable=0    failed=0    skipped=1
rescued=0    ignored=0
```

Step 5: Executing the playbook
(with roles)

```
PLAY [Install and extract Nagios Core] *************************************

TASK [Gathering Facts] *****************************************************
[WARNING]: Platform linux on host server1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [nagios : Update local package index on Ubuntu] ***********************
changed: [server1]

TASK [nagios : Update local package index on CentOS] ***********************
skipping: [server1]

TASK [nagios : Get latest Nagios Core release details] *********************
ok: [server1]

TASK [nagios : Download Nagios Core tarball] *******************************
ok: [server1]

TASK [nagios : Extract Nagios Core tarball] ********************************
ok: [server1]

PLAY RECAP *****************************************************************
server1                    : ok=5    changed=1    unreachable=0    failed=0    sk
ipped=1    rescued=0    ignored=0
```

I ran the playbook with ansible-playbook -i inventory.yaml install_nagios.yaml, and the output showed the Ubuntu task as changed, the CentOS task as skipped, and the remaining tasks as ok, confirming correct OS detection and execution. The play recap showed failed=0, meaning the entire role completed successfully on the target host.

Step 6: Verification

```
kenzo@workstation:~$ ls -la /tmp | grep nagios
drwxrwxr-x 22 root  root     4096 Aug  5 22:54 nagios-4.5.14
-rw-r--r--  1 root  root  2590262 Sep 19 22:17 nagioscore.tar.gz
```

I checked /tmp using ls -la /tmp | grep nagios and confirmed both nagioscore.tar.gz and the extracted nagios-4.5.14 folder were present.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of having an availability monitoring tool?

An availability monitoring tool continuously tracks our servers, networks, and applications to instantly alert the team when a service fails, preventing prolonged outages. It also checks important system details and automatically fixes issues to keep things running smoothly.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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